Infosys Springboard Virtual Internship 6.0 Completion Report

Team Details:

Batch Number: 13

Start date:2-2-2026

Name:Kusum Jha

Internship Duration: 8 Weeks

1. Project Title

NeuroFleetX – AI-Driven Urban Mobility Optimization System

2. Project Objective

The main objective of this project was to design and develop an AI-based system that can improve urban transportation by reducing traffic congestion and increasing fleet efficiency. The system focuses on optimizing routes, predicting vehicle maintenance, and enhancing the overall user experience.

The main goal was to understand how real-world systems combine AI, backend, and frontend technologies to solve practical problems like traffic and mobility.

3. Project description in detail

NeuroFleetX is an AI-powered platform designed to improve urban transportation by integrating real-time data, machine learning concepts, and intelligent decision-making.

The system consists of multiple modules including authentication, fleet management, route optimization, predictive maintenance, and user booking. The backend is developed using Spring Boot, while the frontend is built using React to provide an interactive user interface. Data is managed using MySQL, and AI functionalities are implemented using Python-based logic.

The platform uses simulated vehicle telemetry data such as location, speed, and fuel or battery levels. Route optimization is achieved using shortest path algorithms combined with traffic-based conditions. Predictive maintenance alerts are generated based on vehicle health data trends.

This system can be applied in real-world scenarios such as smart cities, logistics, ride-sharing platforms, and electric vehicle fleet management to improve efficiency and sustainability.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Project understanding, requirement analysis	Finalized project idea and modules
Week 2	UI design and authentication setup	Designed login/register screens
Week 3	Backend setup and database design	Implemented APIs and schema
Week 4	Fleet module development	Vehicle CRUD and telemetry simulation
Week 5	Route optimization integration	Map APIs and route logic implemented
Week 6	Predictive maintenance module	Health analytics dashboard created
Week 7	Booking system and recommendations	Customer UI and AI logic completed
Week 8	Testing, optimization, documentation	Final deployment and report prepared

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Initial planning	Week 1
Prototype/First Draft	Basic UI and backend setup	Week 3
Mid-Term Review	Core modules completed	Week 5
Final Submission	Fully integrated system	Week 8
Presentation	Final demo and explanation	Week 8

5b. Project execution details

The project was executed in a structured and modular manner. It began with requirement analysis and system design, followed by implementation of individual modules.

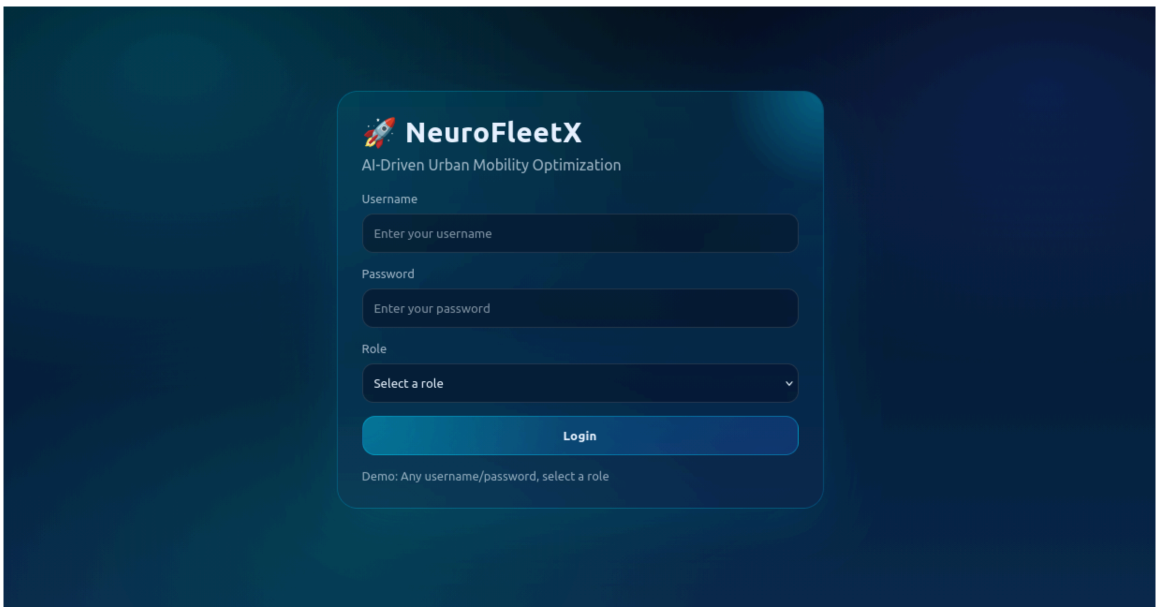
Authentication and dashboard interfaces were developed to establish the system flow. The fleet management module was built using simulated vehicle data. Route optimization was implemented by integrating map APIs and applying shortest path algorithms.

The predictive maintenance module analyzes vehicle data to generate alerts, while the booking system allows users to interact with the platform and receive recommendations.

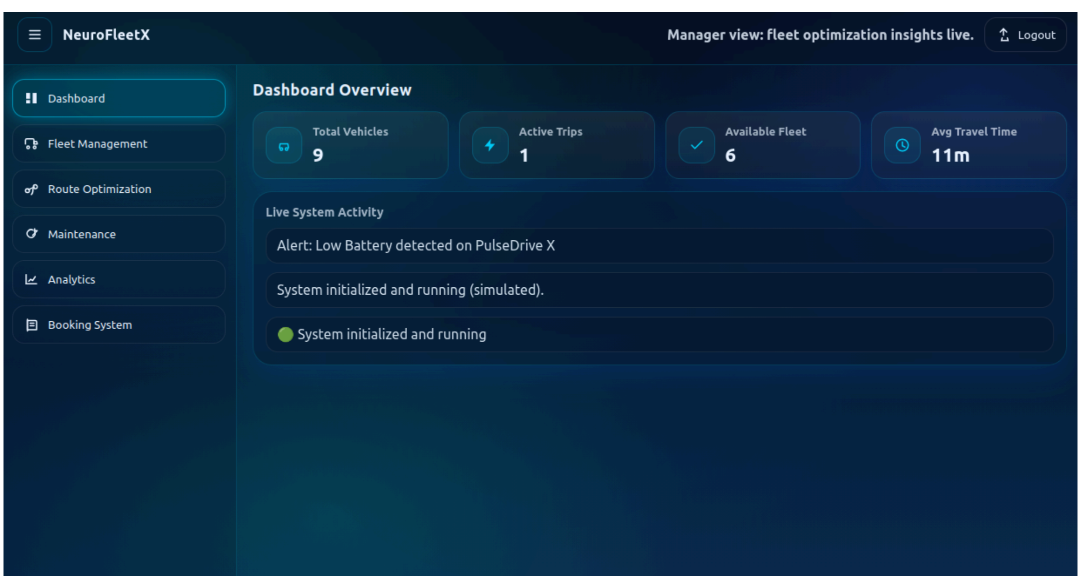
Continuous testing and debugging were performed to ensure smooth integration and system performance.

6. Snapshots / Screenshots

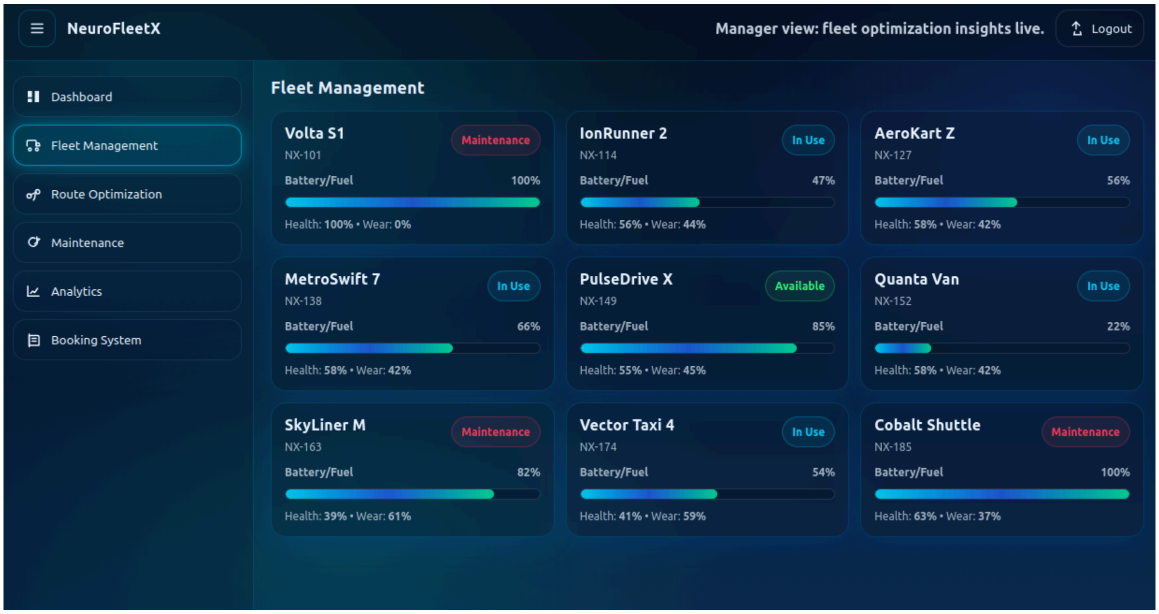
Login UI



Dashboard



Fleet cards



Map with routes

Analytics charts

7. Challenges Faced

Several challenges were encountered during the development of the project:

- Complexity in integrating frontend, backend, and AI components
- Difficulty in working with map APIs and route optimization logic
- Creating realistic simulated data for vehicle telemetry
- Debugging issues during system integration

These challenges were resolved through iterative development, testing, and continuous learning.

8. Learnings & Skills Acquired

This internship helped in gaining both technical and soft skills:

- Technical Skills:
    - Spring Boot & REST APIs
    - React/Angular frontend development
    - Database design using MySQL
    - AI concepts and predictive analytics
    - API integration (Maps, WebSockets)
- Soft Skills:
    - Team collaboration
    - Problem-solving

- Time management
- Project planning

9. Testimonials from team

Working on NeuroFleetX was a highly enriching experience. The project helped me understand real-world applications of AI in smart mobility. I earned how to work as a team, divide responsibilities, and deliver a complete product within deadlines. The internship improved our confidence and technical abilities significantly.

10. Conclusion

The NeuroFleetX project demonstrates how AI and full-stack development can be used to solve real-world problems in urban mobility. It provides an efficient solution for reducing traffic congestion and improving fleet operations.

This project provided valuable practical exposure and strengthened understanding of modern technologies and system integration.

11. Acknowledgements

I would like to express our sincere gratitude to Infosys Springboard for providing this valuable internship opportunity. I also thank our mentor for their continuous guidance and support.